the form of the Hamiltonian, equivalent to a simplified 1DVBE (with viable relaxed compressibility constraint or constant density in solely 1D)

Our SPE, equivalent to 1DVBE is as follows:

$$i\hbar\partial/\partial t(|\psi>) = [(1 - i\frac{2m\nu}{\hbar})\frac{P^2}{2m} - \frac{\hbar^2}{4m\rho_0^2}(\frac{1}{2}|s_x|^2) + \frac{i\hbar\nu|\psi_x|^2}{\rho_0}]\mathbf{I}|\psi> + \vec{\sigma}.\vec{B}|\psi> \quad (1)$$

where $|\psi>$ is not a complex but a quaternionic function. What is a quaternion? Basically it is like a tree with two branches on each of which a complex function sits, but the branches do not have a way to each other (mathematically these two complex parts are on different bases)...Anyway: $\psi(x,t) = a(x,t) + ib(x,t) + jc(x,t) + kd(x,t)$...these a,b,c,d are real functions in this form.

$$i\hbar\partial/\partial t(|\psi>) = [(1 - i\frac{2m\nu}{\hbar})\frac{P^2}{2m} - \frac{\hbar^2}{4m\rho_0^2}(\frac{1}{2}|s_x|^2) + \frac{i\hbar\nu|\psi_x|^2}{\rho_0}]\mathbf{I}|\psi> + \vec{\sigma}.\vec{B}|\psi> \quad (2)$$

Where the P operator has the form of $-\hbar^2\partial_x^2$ for our ultimate use cases. The IHSE portion of this highly nonlinear SPE is as follows:

$$H_1|\psi>= [\frac{P^2}{2m} - \frac{\hbar^2}{8m\rho_0^2}|s_x|^2]\mathbf{I}|\psi> \quad (3)$$

If you note the mainly referenced paper, there is another term for potential V(x)...Here we do not have it...the same applies to pressure, so now barring those we have eq.31 in the main reference...

The Hamiltonian $\frac{P^2}{2m} - \frac{\hbar^2}{4m\rho_0^2}(\frac{1}{2}|s_x|^2)$ is the ISF for which a circuit was proposed in the main reference; however, in that circuit naturally pressure was calculated...We don't want that...it's spurious...so take the latest circuit Jupyter notebook for the ISF and eliminate the presence of pressure in that circuit!

But then comes the more challenging part! There are other terms to the SPE, not present in the IHSE:

$$H_2|\psi >= [i\hbar\nu(\partial_x^2 + \frac{|\psi_x|^2}{\rho_0})]\mathbf{I} + \vec{\sigma}.\vec{B}]|\psi > \tag{4}$$

Where:

$\sigma = \sigma^j e_j = [[0,1],[1,0]]e_1 + [[0,-i],[i,0]]e_2 + [[1,0],[0,-1]]e_3$

$$B = \frac{\hbar^2}{4m\rho_0}s_{xx} + \frac{\hbar^2}{4m\rho_0^3}[\tilde{s}|\tilde{s}_x|^2] - \frac{2m\nu|\psi_x|^2 u}{|\tilde{s}_x|^2} \tag{5}$$

u must be rewritten based on ψ:

$$u = \frac{-i\hbar}{m\rho_0}(\bar{\psi}\partial_x\psi) \tag{6}$$

$$s = \bar{\psi}\sigma\psi \tag{7}$$

$$\tilde{s} = \bar{\psi}i\psi \tag{8}$$

So $|A|^2$ for the quaternion means $\bar{A}A$, where $\bar{A}$ can be translated to $[A_+^* \; A_-^*]$, where A_+^* and A_-^* are complex conjugates of functions

With regard to H_2:
Note the first term is non-Hermitian & linear in ψ
the second term non-Hermitian & non-linear in ψ
the Third one, Hermitian & non-linear in ψ

At the end of the day, what we want is of course $U = e^{-iH_2}$, attached to the IHSE Qiskit circuit, wherein the pressure-related ingredients have been removed...
This is possibly achievable via Variational Quantum Algorithm for Nonlinear Problems